

Assignment

Objective

- We have a folder that contains the audio files, i.e. .wav files. These audio files are of multiple categories like dog, chirping-birds, vacuum-cleaner, thunderstorm, crow, can-opening like this many categories of audio file.
- Our model should classify which category it is so it is a multi-class classification problem.
- I need to construct 2 models
 - CNN model
 - Transformer model It has the <cls> token also.
- Also in the dataset I have the .csv file the columns of that csv file is
 - file name [i.e. audio file name]
 - Category
 - Category-label [this is more important for us]

Answer)

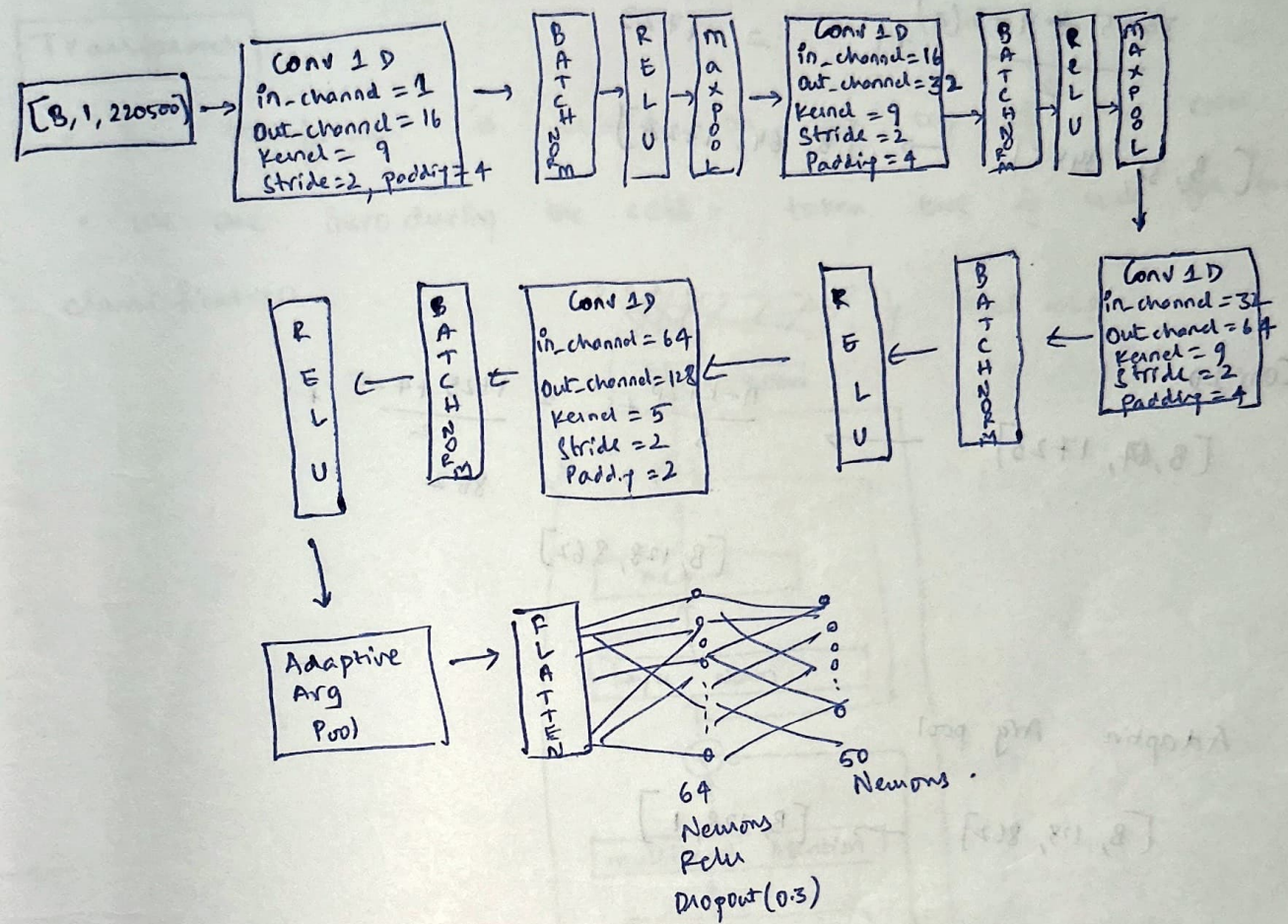
Preprocessing

- each audio file is .wav format
- each audio file duration is 5 sec
- Audio channels = 1 (i.e. Mono)
- Sample rate = $44.1 \text{ kHz} = 44100 \text{ Hz}$
- Bits per Sample = 16.
- There are 50 categories of audio there exist here.
- So we are using the torch audio to get the waveform
- $44100 \times 5 \text{ sec} = 220500$
and 1 channel so the dataset class returns the $[B, 1, 220500]$ and also the category label

CNN model

→ It is not the 2D CNN It is 1D CNN

Architecture



- the Tensor that I get from the Dataloader is $[B, 1, 220500]$

- Conv 1D
$$\frac{n-k+2p}{s} + 1 = \frac{220500 - 9 + 2(4)}{2} + 1 = 110250$$

So $[B, 1, 220500] \xrightarrow{\text{Conv 1D}} [B, 16, 110250]$

Also Batch Norm (16), ReLU Applied

- $\text{Maxpool 1D}(4) \Rightarrow [B, 16, 110250] \rightarrow [B, 16, 27562]$

- Conv 1D
$$\frac{n-k+2p}{s} + 1 = \frac{27562 - 9 + 2(4)}{2} + 1 = 13781$$

$[B, 16, 27562] \rightarrow [B, 32, 13781]$

• max pool

$$[B, 32, 13781] \rightarrow [B, 32, 3445]$$

• Conv 1D

$$\frac{3445 + 9 + 2(4)}{2} + 1 = 1723$$

$$[B, 32, 3445] \rightarrow [B, 64, 1723]$$

• Conv 1D

$$[B, 64, 1723]$$

$$\frac{n-k+2p}{s} + 1 = \frac{1723 + 4 - 5}{2} + 1 = 862$$

$$[B, 128, 862]$$

• Adaptive Arg pool

$$[B, 128, 862] \rightarrow [B, 128, 1]$$

• Flatten

$$[B, 128]$$

$$[B, 128] \cdot [128, 64] = [B, 64] \quad \left. \vphantom{[B, 128] \cdot [128, 64]} \right\} \text{fully connected}$$

$$[B, 64] \cdot [64, 50] = [B, 50]$$

Loss

multi class classification so categorical cross entropy

$$\text{Loss} = - \sum_i y_i \log \hat{y}_i$$

Parameters to train the model

epochs = 150

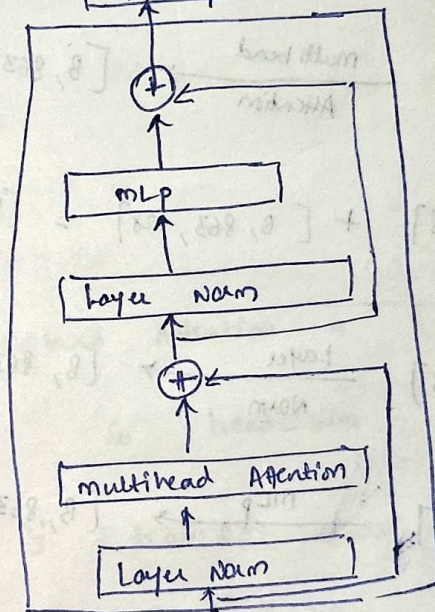
num_classes = 50

batch_size = 32

Transformer

- This transformer is created on the top of the CNN
- we are introducing the $\langle \text{cls} \rangle$ token that is used for the classification

final mlp



BATCH NORM

ReLU

Conv 2D
in-channel=64
Out-channel=128
Kernel=5
Stride=2
Padding=2

ReLU

BATCH NORM

Conv 1D
in-channel=32
Out-channel=64
Kernel=9
Stride=2 padding=4

Max Pool

ReLU

[B, 1, 220500]

Conv 1D
in-channel=1
Out-channel=16
Kernel=9
Stride=2 padding=4

BATCH NORM

ReLU

Max Pool

Conv 2D
in-channel=16
Out-channel=32
Kernel=9
Stride=2 padding=4

BATCH NORM

• From the CNN we get the output as $[B, 128, 862]$

• $\text{Transpose}(1, 2) = [B, 862, 128]$

• Concat class token $[B, 863, 128]$

• Add the position embedding info to the token embedding

$$[B, 863, 128]$$

• pass through the Transformer

$$[B, 863, 128] \xrightarrow[\text{NOAM}]{\text{Layer}} [B, 863, 128]$$

$$[B, 863, 128] \xrightarrow[\text{Attention}]{\text{multi head}} [B, 863, 128]$$

$$[B, 863, 128] + [B, 863, 128] = [B, 863, 128]$$

$$[B, 863, 128] \xrightarrow[\text{NOAM}]{\text{Layer}} [B, 863, 128]$$

$$[B, 863, 128] \xrightarrow{\text{MLP}} [B, 863, 128]$$

$$[B, 863, 128] + [B, 863, 128] = [B, 863, 128]$$

$$[B, 863, 128] \xrightarrow[\text{Layer NOAM}]{\text{final}} [B, 863, 128]$$

cls token $[:, 0, :]$ token is taken

pass through the final MLP

$$[B, 1, 128] \cdot [128, 256] = [B, 1, 256]$$

$$[B, 1, 256] \xrightarrow{\text{GELU}(\cdot)} [B, 1, 256]$$

$$[B, 1, 256] \cdot [B, 256, 50] = [B, 1, 50]$$

• GELU()

$x \rightarrow \phi(x)$ where $\phi(x)$ is the Cumulative probability distribution of the Standard Normal distribution.

Parameters of Transformer

batch-size = 32

num-classes = 50

epochs = 100

embed-dim = 128

num-transformer-blocks = 2

dropout = 0.1

num-tokens = 862

num-heads = 4

Multihead Attention

→ the input to multihead Attention is $[B, 863, 128]$

→ Also num-heads = 4. So head-dim = $128 // 4 = 32$

→ Here 3 Constructed 3 trainable matrices W_Q, W_K, W_V and dimensions of them are $[128, 128]$ all the 3 are $[128, 128]$

$$\rightarrow Q = X \cdot W_Q = [B, 863, 128] \cdot [128, 128] = [B, 863, 128]$$

$$K = X \cdot W_K = [B, 863, 128] \cdot [128, 128] = [B, 863, 128]$$

$$V = X \cdot W_V = [B, 863, 128] \cdot [128, 128] = [B, 863, 128]$$

→ Q Reshaped to $[B, 4, 863, 32]$ i.e. $[B, \text{num-heads}, \text{num_tokens} + 1, \text{head_dim}]$
 <cls> also involved

→ K Reshaped to $[B, 4, 863, 32]$ [why 863 because <cls> token also there]

→ V Reshaped to $[B, 4, 863, 32]$

→ Now formula

$$\text{softmax} \left(\frac{\Theta \cdot K^T}{\sqrt{d_k}} \right) \cdot V$$

→ So $\Theta \cdot K^T$

$$\begin{bmatrix} B, 863 \end{bmatrix} \cdot \begin{bmatrix} B, 4, 863, 32 \end{bmatrix} \cdot \begin{bmatrix} B, 4, 32, 863 \end{bmatrix} \\ = \begin{bmatrix} B, 4, 863, 863 \end{bmatrix}$$

→ d_k is Nothing But the head-dim = 32
 $\sqrt{32}$

the shape will be $\begin{bmatrix} B, 4, 863, 863 \end{bmatrix}$

→ Here Current token has the Access of the future tokens this is Normal multihead Attention Not the Causal Attention

→ Now softmax is Applied $\begin{bmatrix} B, 4, 863, 863 \end{bmatrix}$

$$\begin{bmatrix} B, 4, 863, 863 \end{bmatrix} \cdot \begin{bmatrix} B, 4, 863, 32 \end{bmatrix} \quad \text{multiply with } V \\ = \begin{bmatrix} B, 4, 863, 32 \end{bmatrix}$$

$$\rightarrow \text{transpose at } (1,2) \quad \begin{bmatrix} B, 863, 4, 32 \end{bmatrix}$$

→ then Convert to $\begin{bmatrix} B, 863, 128 \end{bmatrix}$

→ Now Final projection matrix the shape of

$$\text{matrix is } \begin{bmatrix} B, 863, 128 \end{bmatrix}$$

$$\text{So } \begin{bmatrix} B, 863, 128 \end{bmatrix} \cdot \begin{bmatrix} 128, 128 \end{bmatrix} = \begin{bmatrix} B, 863, 128 \end{bmatrix}$$

MLP inside the transformer

→ the input to MLP is $[B, 863, 128]$

→ the hidden layer of MLP contain 4 times neurons of embedding-dim i.e 128 i.e 512 Neurons

So the weight matrix is $[128, 512]$

→ So $[B, 863, 128] \cdot [128, 512] = [B, 863, 512]$

→ Now GELU() activation is Applied

→ then again projected to the same

$[B, 863, 512] \cdot [512, 128] = [B, 863, 128]$

✱ As usual Residual Connection is Applied Here.

• This is entire Comparison with CNN and Transformer.

